

awk Cheat Sheet

Work with text in columns: pick fields, filter rows, count and add things up.

Using it

```
awk 'pattern { action }' file ← run action on every line that matches pattern
command | awk '{ print $1 }' ← work on another command's output
```

awk reads one line at a time and splits it into **fields** **\$1**, **\$2**... (by default at any run of spaces or tabs). No pattern = every line. No action = print the line.

Option	What it does	Option	What it does
-F ':'	Splits fields at : instead of spaces	-v n=5	Sets a variable before starting
-F ','	Splits at commas (simple CSV)	-f prog.awk	Reads the program from a file

Fields and built-in variables

```
$ echo 'alice 42 london' | awk '{ print $3, $1 }'
london alice ← $1 = alice · $2 = 42 · $3 = london · $0 = the whole line
```

Name	Meaning	Name	Meaning
\$0	The whole line	NR	Line number (counting across all files)
\$1, \$2	Field 1, field 2...	FNR	Line number within the current file
NF	Number of fields on this line	FS / OFS	Input / output field separator
\$NF	The last field (\$(NF-1) = second to last)	FILENAME	Name of the file being read

Patterns (which lines; combine with && || !)

Pattern	Lines it picks	Pattern	Lines it picks
/error/	Lines containing error	NR == 1	Only the first line
!/error/	Lines not containing error	NR > 1	Every line except the first (skip a header)
\$3 > 100	Field 3 is greater than 100	NR==5, NR==10	Lines 5 to 10
\$1 == "root"	Field 1 is exactly root	BEGIN	Runs once before any input
\$2 ~ /^a/	Field 2 matches a pattern (!~ = doesn't)	END	Runs once after all input

Actions (what to do)

Action	Does	Action	Does
print \$1, \$3	Prints fields with a space between	length(\$1)	Length of a field (or of the line)
print \$1 \$3	Prints fields joined with no space	toupper(\$1)	Uppercase (to lower = lowercase)
printf	printf "%-10s %5d\n", \$1, \$2	substr(s,2,3)	3 characters of s , from position 2
sum += \$3	Adds field 3 to a running total	gsub(/a/, "b")	Replaces every a with b in the line
count[\$1]++	Counts lines per value of field 1	if (x) { ... }	Does something only when true

Handy recipes

Command	Does
awk -F: '{ print \$1 }' /etc/passwd	Lists every user name
awk '{ s += \$5 } END { print s }' file	Adds up column 5
awk '{ s += \$2 } END { print s/NR }' file	Averages column 2
awk 'NR > 1' data.csv	Drops the header line
awk '{ c[\$1]++ } END { for (k in c) print c[k], k }' log	Counts lines per value of field 1 (e.g. per IP)
awk '!seen[\$0]++' file	Removes duplicate lines, keeping the original order
df -h awk '\$5+0 > 80'	Shows disks more than 80% full
awk -F, -v OFS=, '{ print \$2, \$1 }' f.csv	Swaps the first two CSV columns

Remember: wrap the program in **single quotes** so the shell doesn't touch **\$1**. Inside the program, text goes in **double quotes**: **awk '\$1 == "root"'**.